卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-EC-0406 30-B003701S-D0406

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图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

电气 专业 第 04 卷 第 06 册
Electrical SPECIALITY VOLUME 04 BOOK 06

卷册名称 户外电缆设施图--直流场与换流区域
Volume name Outdoor Cable Facility-DC Yard and Converter Area

批 准 APPROVED BY	2019.01.24
审 核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	2019.01.24
图纸 7 张 DRAWING 7 NOS.	打印件 本 PRINTED COPIES NOS.

序号 No.	图 号 DRAWING NO.	文件名称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	30-B003701S-D0406-001	电气设计施工说明 Electrical design construction instructions	A	1		
2	30-B003701S-D0406-002	户外电缆沟及埋管布置图 Layout drawing of outdoor cable trench and buried pipe	A	1		
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4	30-B003701S-D0406-004	电缆过渡支架加工示意图 Detail drawing of cable transition bracket	A	1		
5	30-B003701S-D0406-005	户外光缆槽盒布置图 Layout drawing of outdoor optical cable box	A	1		
6	30-B003701S-D0406-006	雨水泵站电缆设施示意图 Detail drawing of cable facilities for rainwater pump station	A	1		
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注:

电气设计施工说明

Electrical design construction instructions

一、项目概况：

项目名称：巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程/拉合尔±660kV换流站

建设单位：国家电网/中国电力技术装备有限公司

本次出图范围为直流场、换流变区域及站前区户外电缆设施图，主要包括：直流场、换流变区域户外电缆沟内电缆支架布置和制作；全站户外电缆沟光缆槽盒布置等。

二、施工说明

1. 电缆支架采用热镀锌角钢制作：电缆立柱通过膨胀螺栓与电缆沟壁固定；电缆格架与立柱可靠焊接.
2. 电缆支架间隔为0.8m，在电缆沟两侧交错布置。同时在电缆沟交叉处设置电缆过渡托架，格架端部加装护齿以保护电缆不被损伤。
3. 电缆沟内的所有电缆必须敷设在支架上。电力电缆、控制电缆、光缆等在支架上的排列顺序为（从下而上）：K0（35kV、10kV电缆）、K1（400V动力电缆）、K2控制和信号电缆、K3（电流电压测量信号电缆）、K4（直流电流电压测量信号电缆）、K5（光缆）、K6（通信电缆）。光缆槽盒应布置在第一层支架上（从上到下）。
4. 每个电缆沟支架均应可靠接地，通长扁钢与每个电缆支架通过螺栓连接，并与土建预留的每个接地引上线连接。接地引上线要求及材料量参见D0403及D0411卷册。
5. 电缆应按照本卷册中规定的路径敷设；如无特殊规定则应以最短路径敷设，若局部电缆过于密集，可适当迂回。
6. 本期的电缆敷设应按远景敷设要求考虑,并为远景电缆敷设预留空间，尽量避免电缆互相交叉。
7. 各类电缆的转弯半径应符合相关规范的要求或电缆厂家的特殊规定。
8. 单芯电力电缆进出电缆沟或墙体时应穿PVC/UPVC管，且绑定物不能形成闭合磁路。管中电缆不宜过多，一般管径应不小于电缆束外径的1.5倍。
9. 光纤在电缆沟中时敷设在光纤槽盒内，光缆槽盒固定在电缆支架上。
10. 为方便场地中电缆进出电缆沟，电缆沟壁上已预留套管或已在电缆沟旁边设置电缆井，施工单位敷设电缆时应注意套管或井的位置，尽量避免重新开孔。
11. 消防、报警、应急照明、断路器操作直流电源等重要回路应采用防火隔板进行分隔.
12. 计算机监控、双重化继电保护、保安电源等双回路应尽量采用不同路径敷设，当敷设于同一通道时，应敷设于通道两侧或同一侧支架的不同层。

三、其他

1. 直流场、换流变区域及站前区户外电缆设施由电气安装A包负责供货及施工。
2. 电缆设施施工存在但不限于以下安全风险：(1)机械伤害；(2)机械触电;(3)高处坠落；(4)物体打击，触电火灾等。施工时应严格按照国家、地方及国家电网公司相关安全生产法律法规执行，保证施工安全。
3. 施工中除满足本说明外，还应满足相关施工验收规范、招标技术文件及相关规程要求，包括但不限于：GB 50217-2018电力工程电缆设计规范。

First: Project overview

Project name：±660kV HVDC Project from Matiari to Lahore in Pakistan/±660kV Lahore Converter Station

Development organization：State Grid / China Electric Power Equipment And Technology CO .TD

Design scope: This volume is an outdoor cable facility diagram of DC Yard, Converter Area and Station Front Area, mainly including: the cable bracket layout and production in outdoor cable trench ; the cable box layout in outdoor cable trench of the whole station , etc.

Second: Construction Instructions

1. The cable bracket is made of hot-dip galvanized angle steel: the column is fixed with cable trench wall through expansion bolts;by the expansion bolt; the cable grid is reliably welded with the column;
2. The cable bracket spacing is 0.8m , staggered arrangement on both sides of cable trench. Cable transition brackets are provided at the intersections of the cable trenches,and protective teeth are added at the ends of the grids to protect the cables from being damaged.
3. All cables in the cable trench must be laid on the bracket. Power cables, control cables and optical cables are arranged in the order of (from bottom to top): K0 (35kV, 10kV cables), K1 (400V power cables), K2 (control and signal cables), K3 (current and voltage measurement signal cables), K4 (DC current and voltage measurement signal cables), K5 (optical cables), K6 (communication cables). The cable groove box shall be arranged on the first bracket (from top to bottom).
4. Each cable trench support shall be reliably earthed, and the through-length flat steel shall be bolted to each cable support, and shall be connected to each earthing lead reserved by civil engineering. Please refer to volume D0403 and D0411 for grounding lead requirements and material quantity.
5. Cables shall be laid along the routes specified in this volume; If no special requirements cables should be laid in the shortest path, if the local cable is too dense, the path can be appropriate circuitous.
6. The cable laying in the current period shall be considered according to the long-term laying requirement, and reserved space for the future cable, so as to avoid cables crossing each other as far as possible.
7.The turning radius of all types of cables shall meet the requirements of relevant specifications or the special requirements of cable manufacturers.
8. When single-core power cable enters or leaves the cable trench or wall, it shall wear PVC/UPVC pipe, and the binding material shall not form a closed magnetic circuit. The cable in the pipe should not be too much, the general pipe diameter should not be less than 1.5 times the diameter of the cable bundle.
9. Optical cable in trench laying within the cable box, and the cable box is fixed to the cable stand.
10. To facilitate the access to the cable trench in the site, the cable casing has been reserved on the cable trench wall or the cable pit has been set up beside the cable ditch.When laying the cable, the construction unit should pay attention to the cable casing or cable pit and try to avoid re-opening the hole.
11 Important circuits such as fire protection, alarm, emergency lighting, circuit breaker operating dc power supply shall be separated by fire bulkheads.
12 Computer monitoring, double circuit relay protection and security power should be laid in different paths as far as possible. When laying in the same channel, it should be laid on both sides of the channel or different layers of the bracket on the same side.

Third： Else

1. The outdoor cable facility of DC field, converter area and station front area is supplied and constructed by the electrical installation A package.
2. The construction of cable facilities exists but is not limited to the following safety risks:
   (1) mechanical damage; (2) mechanical shock; (3) falling from high places;
   (4) hitting objects, electric shocks, etc.
   Construction should be carried out in strict accordance with the relevant national and local power grid company safety production laws and regulations to ensure construction safety.
3. In addition to this specification, the requirements of relevant construction acceptance specification,bidding technical documents and relevant regulations shall be fulfilled during construction, including but not limited to: GB 50217-2018 code for design of cables of electric engineering.

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE：施工图 Detail Design

M L H V D C - L C S - E C I - D D - E C - 0 4 0 6 - 0 0 1 A

	姓名 Name	日期 Date
设计 Designed	张云鹏	2019.01.20
校核 Checked	李兼明	2019.01.22
审核 Reviewed		
批准 Approved	郑煜	2019.01.24
比例 Scale		图号 DWG NO

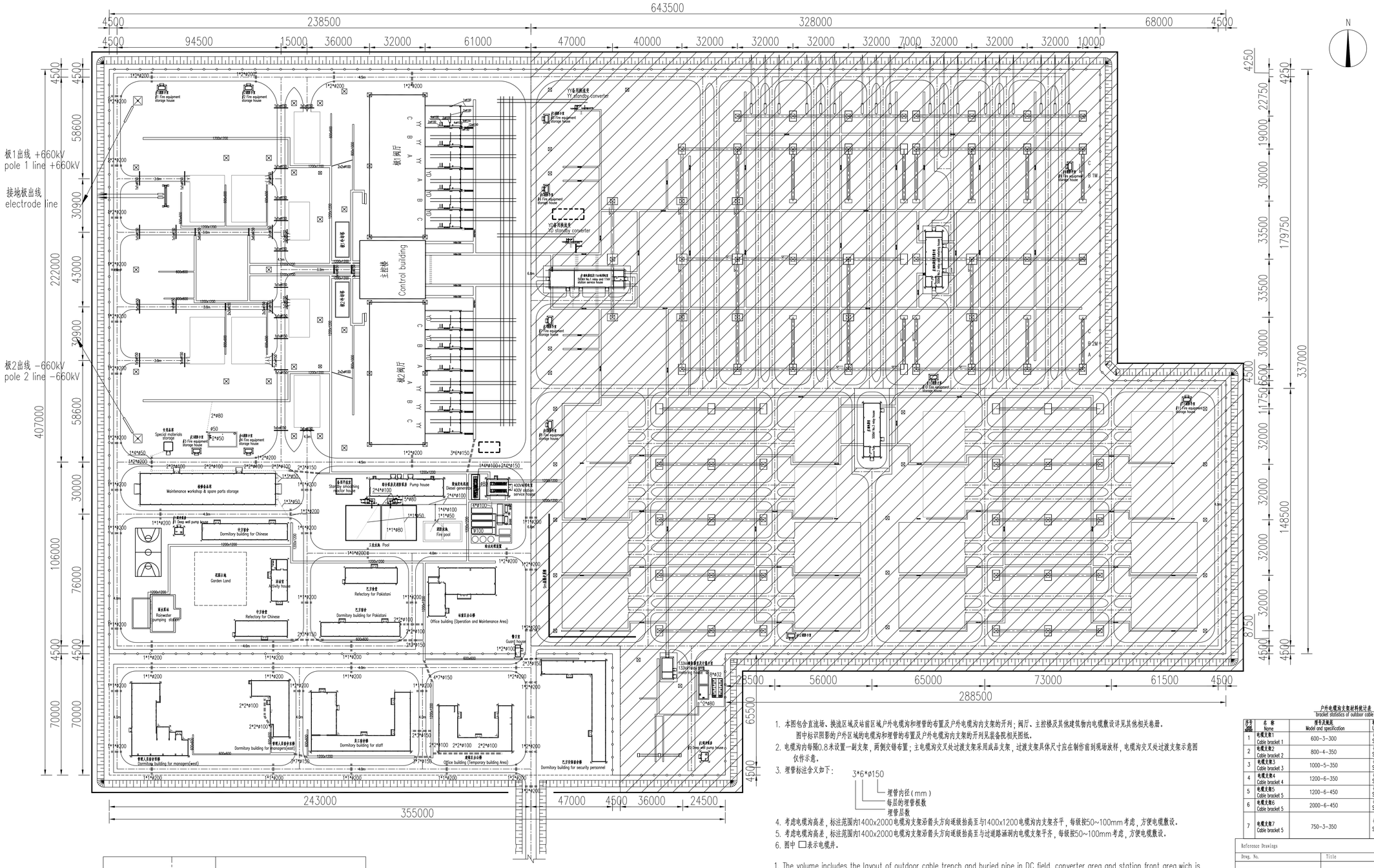
电气设计施工说明

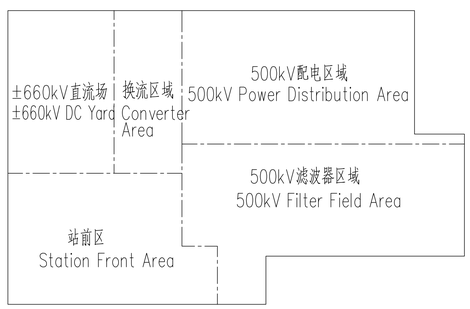
Electrical design construction instructions

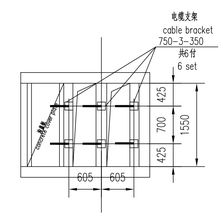
30-B003701S-D0406-001

版本 REV A

297X420







换流变工控柜下电缆桥架布置图

Layout of cable bracket under converter transformer's control cabinet

1. 本图包含直流场、换流区域及站前区域户外电缆沟和埋管的布置及户外电缆沟内支架的开列；阀厅、主控楼及其他建筑物内电缆敷设详见其他相关卷册。
   图中标识阴影的户外区域的电缆沟和埋管的布置及户外电缆沟内支架的开列见装备院相关图纸。
2. 电缆沟内每隔0.8米设置一副支架，两侧交错布置；主电缆沟交叉处过渡支架采用成品支架，过渡支架具体尺寸应在制作前到现场放样，电缆沟交叉处过渡支架示意图仅作示意。
3. 埋管标注含义如下：

   3*6*ø150
   - 埋管内径（mm）
   - 每层的埋管根数
   - 埋管层数

4. 考虑电缆沟高差，标注范围内1400x2000电缆沟支架沿箭头方向逐级抬高至与1400x1200电缆沟内支架齐平，每级按50~100mm考虑，方便电缆敷设。
5. 考虑电缆沟高差，标注范围内1400x2000电缆沟支架沿箭头方向逐级抬高至与过道路涵洞内电缆支架平齐，每级按50~100mm考虑，方便电缆敷设。
6. 图中 □ 表示电缆井。

1. The volume includes the layout of outdoor cable trench and buried pipe in DC field, converter area and station front area,wich is outside the shadow area. Cable brackets of these cable trench are listed in the volume. The cable layout of valve hall, main control building and other buildings is detailed in the relevant volumes of our institute.The layout of outdoor cable trench and buried pipe within the shadow area is detailed in the relevant volumes of CET.
2. A set of bracket shall be set up every 0.8 meters on one side of the cable trench.If brackets shall be set up on both side,it shall be staggered. Finished bracketsshall be taken as transition brackets at the intersection of the cable trench . The schematic diagram of the transition brackets is only schematic,its specific size should be set out on site before production.
3. The meaning of the buried tube is as follows:

   3*6*ø150
   - Inner diameter of buried pipe (mm)
   - The number of tubes buried per layer
   - The layer number of buried pipe

4.Considering the cable trench elevation difference, the cable trench bracket in the labeled range of 1400x2000 cable trench shall be gradually raised up along the arrow direction to be flush with the bracket of 1400x1200 cable trench.Each level shall be considered as 50~100mm, which is convenient for cable laying.
5.Considering the cable trench elevation difference, the cable trench bracket in the labeled range of 1400x2000 cable trench shall be gradually raised up along the arrow direction to be flush with the bracket Inside the road culvert.Each level shall be considered as 50~100mm, which is convenient for cable laying.
6. means the cable well in this picture.

户外电缆沟支架材料统计表
bracket statistics of outdoor cable trench

序号 Item	名称 Name	规格及规范 Model and specification	单位 Unit	数量 Quantity	备注 Remark
1	电缆支架1 Cable bracket 1	600-3-300	副 Set	581	用于600x600电缆沟,单侧安装 For 600x600 cable trench,set on one side
2	电缆支架2 Cable bracket 2	800-4-350	副 Set	110	用于800x800电缆沟,单侧安装 For 800x800 cable trench,set on one side
3	电缆支架3 Cable bracket 3	1000-5-350	副 Set	115	用于800x1000电缆沟,单侧安装 For 800x1000 cable trench,set on one side
4	电缆支架4 Cable bracket 4	1200-6-350	副 Set	3720	用于1200x1200电缆沟,双侧安装 For 1200x1200 cable trench, set on both sides
5	电缆支架5 Cable bracket 5	1200-6-450	副 Set	286	用于1400x1200电缆沟,双侧安装 For 1400x1200 cable trench, set on both sides
6	电缆支架6 Cable bracket 5	2000-6-450	副 Set	612	用于1400x2000电缆沟,双侧安装 For 1400x2000 cable trench, set on both sides
7	电缆支架7 Cable bracket 5	750-3-350	副 Set	72	用于换流变工控柜电缆井 For cable well of converter transformer's Control cabinet

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV换流站工程
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拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - E C - 0 4 0 6 - 0 0 2 A

	姓名 Name	日期 Date
设计 Designed	张东亮	2019.01.20
校核 Checked	傅家明	2019.01.22
审核 Reviewed		
批准 Approved	郑煜	2019.01.24

户外电缆沟及埋管布置图
Layout drawing of outdoor cable trench and buried pipe

比例 Scale: 1:800 | 图号 DWG NO.: 30-B003701S-D0406-002 | 版本 REV.: A

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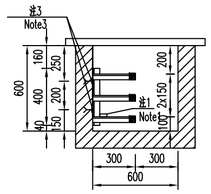
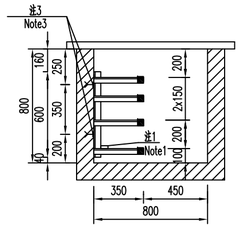


600x600电缆沟
600x600 cable trench

600-3-300
布置在电缆沟单侧，支架间距0.8米布置一副
直流场、站前区
图1

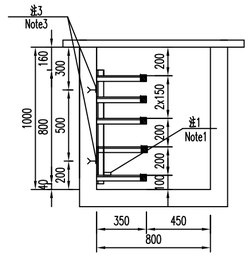
Set on one side of the cable trench, with a interval of 0.8m
Dc field, Station front area
Figure 1



800x800电缆沟
800x800cable trench

800-4-350
布置在电缆沟单侧，支架间距0.8米布置一副
换流变区域
图2

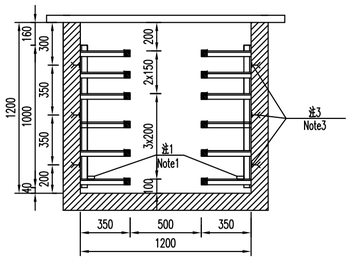
Set on one side of the cable trench, with a interval of 0.8m
Station front area
Figure 2



800x1000电缆沟
800x1000cable trench

1000-5-350
布置在电缆沟单侧，支架间距0.8米布置一副
直流场
图3

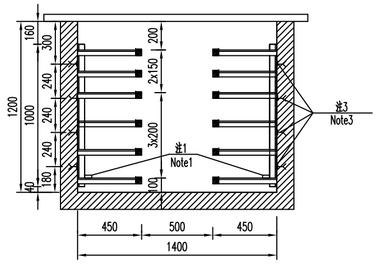
Set on one side of the cable trench, with a interval of 0.8m
Dc field
Figure 3



1200x1200电缆沟
1200x1200 cable trench

1200-6-350
布置在电缆沟两侧，支架间距0.8米布置一副
两侧支架应错位布置
直流场、站前区
图4

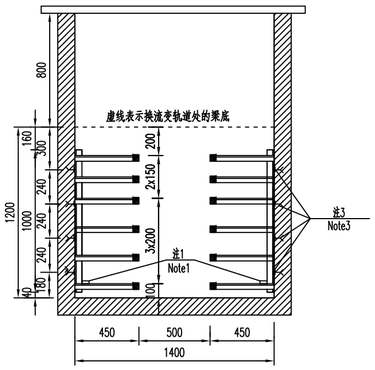
Set on both sides of the cable trench, with a interval of 0.8m
the brackets on both sides should be staggered
Dc field, Station front area
Figure 4



1400x1200电缆沟
1400x1200 cable trench

1200-6-350
布置在电缆沟两侧，支架间距0.8米布置一副
两侧支架应错位布置
换流变区域
图5

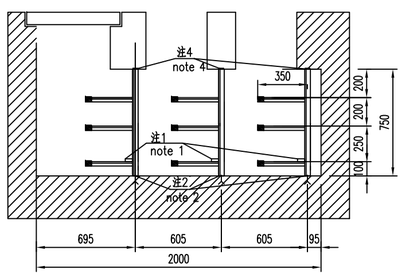
Set on both sides of the cable trench, with a interval of 0.8m
the brackets on both sides should be staggered
Converter area
Figure 5



1400x2000电缆沟
1400x2000 cable trench

2000-6-450
布置在电缆沟两侧，两侧支架应错位布置
支架间距0.8米布置一副
换流变区域
图6

Set on both sides of the cable trench, with a interval of 0.8m
the brackets on both sides should be staggered
Converter area
Figure 6



换流变工控柜下电缆支架
Cable bracket used for converter transformer's Control cabinet

750-3-350
布置在工控柜下方，布置平面图见"D0406-002"图
换流变区域
图7

Set under converter transformer's Control cabinet
layout can be seen on"D0406-002"
Converter area
Figure 7

注：

1、该处敷设-50x5通长扁钢，材料量开列在D0403卷册内，通长扁钢用热镀锌钢制M12螺栓与每个电缆支架固定,用于电缆支架接地；

2、标注处将过渡钢板利用M12热镀锌钢制内膨胀螺栓固定在电缆沟底，角钢立柱与钢板焊接固定；

3、标注处使用M12热镀锌钢制内膨胀螺栓将角钢立柱与电缆沟固定，连接处支架建议开长孔；

4、标注处角钢立柱与土建预埋件焊接；

5、角钢支架加工时应保证横架、立柱表面光洁不受破坏；圆角、边缘等处光滑无毛刺和残渣；支架端部应当装设保护套，保护套颜色应全站保持一致。

6、施工中还应满足相关的施工验收规范。

Note：

1、A 50x5 long flat steel is seted in this place and its material is listed in the Volume D0403 .The long flat steel shall be fixed with each cable bracket using M12 ,hot dip galvanizing bolt for cable bracket grounding.

2、In this place, the transition steel plate is fixed on the bottom of the cable trench with M12 ,hot dip galvanizing expansion bolt. Then column is welded with steel plate；

3、In this place, the angle steel column is fixed to the cable trench by M12 hot dip galvanizing expansion bolt. Long hole is recommended for the joint bracket.

4、In this place,the angle steel column is welded with civil engineeringembedded parts.

5、When processing the angle steel bracket, ensure that the surface of the horizontal frame and the column are not damaged; the rounded corners and edges are smooth and free of burrs and debris; A protective cover shall be installed at the end of the bracket, and the color of the protective cover shall be consistent throughout the station.

6、The construction should also meet the related construction acceptance specifications.

支架制作表
Bracket making table

序号 Serial number	名称 Name	型号及规格 Model and specification	单位 Unit	数量 Quantity	单重/kg Piece weight/kg	合计/kg Total	总重/kg Gross weight/kg	备注 Remark
1	电缆支架1 Cable bracket 1	600-3-300	套 Set	1			3.88	用于600x600电缆沟 For 600x600 cable trench
	角钢立柱 Angle steel column	L50x50x5，L=400mm	根 Pcs	1	1.6	1.6		
	角钢悬架 Angle steel bracket	L40x40x4，L=300mm	套 Set	3	0.76	2.28		
	膨胀螺栓 Expansion bolt	M12x100，热镀锌 M12x100, hot dip galvanizing	套 Set	2				
	螺栓 bolt	M12，热镀锌 M12, hot dip galvanizing	套 Set	1				
2	电缆支架2 Cable bracket 1	800-4-350	套 Set	1			5.96	用于800x800电缆沟 For 800x800 cable trench
	角钢立柱 Angle steel column	L50x50x5，L=600mm	根 Pcs	1	2.4	2.4		
	角钢悬架 Angle steel bracket	L40x40x4，L=350mm	套 Set	4	0.89	3.56		
	膨胀螺栓 Expansion bolt	M12x100，热镀锌 M12x100, hot dip galvanizing	套 Set	2				
	螺栓 bolt	M12，热镀锌 M12, hot dip galvanizing	套 Set	1				
3	电缆支架3 Cable bracket 1	1000-5-350	套 Set	1			7.65	用于800x1000电缆沟 For 800x1000 cable trench
	角钢立柱 Angle steel column	L50x50x5，L=800mm	根 Pcs	1	3.2	3.2		
	角钢悬架 Angle steel bracket	L40x40x4，L=350mm	套 Set	5	0.89	4.45		
	膨胀螺栓 Expansion bolt	M12x100，热镀锌 M12x100, hot dip galvanizing	套 Set	2				
	螺栓 bolt	M12，热镀锌 M12, hot dip galvanizing	套 Set	1				
4	电缆支架4 Cable bracket 1	1200-6-350	套 Set	1			9.34	用于1200x1200电缆沟 For 1200x1200 cable trench
	角钢立柱 Angle steel column	L50x50x5，L=1000mm	根 Pcs	1	4	4		
	角钢悬架 Angle steel bracket	L40x40x4，L=350mm	套 Set	6	0.89	5.34		
	膨胀螺栓 Expansion bolt	M12x100，热镀锌 M12x100, hot dip galvanizing	套 Set	3				
	螺栓 bolt	M12，热镀锌 M12, hot dip galvanizing	套 Set	1				
5	电缆支架5 Cable bracket 1	1200-6-450	套 Set	1			14.8	用于1400x1200电缆沟 For 1400x1200 cable trench
	角钢立柱 Angle steel column	L50x50x5，L=1000mm	根 Pcs	1	4	4		
	角钢悬架 Angle steel bracket	L50x50x5，L=450mm	套 Set	6	1.8	10.8		
	膨胀螺栓 Expansion bolt	M12x100，热镀锌 M12x100, hot dip galvanizing	套 Set	4				
	螺栓 bolt	M12，热镀锌 M12, hot dip galvanizing	套 Set	1				
6	电缆支架6 Cable bracket 1	2000-6-450	套 Set	1			14.8	用于1400x2000电缆沟 For 1400x2000 cable trench
	角钢立柱 Angle steel column	L50x50x5，L=1000mm	根 Pcs	1	4	4		
	角钢悬架 Angle steel bracket	L50x50x5，L=450mm	套 Set	6	1.8	10.8		
	膨胀螺栓 Expansion bolt	M12x100，热镀锌 M12x100, hot dip galvanizing	套 Set	4				
	螺栓 bolt	M12，热镀锌 M12, hot dip galvanizing	套 Set	1				
7	电缆支架7 Cable bracket 1	750-3-350	套 Set	1			6.17	用于换流变工控柜电缆井 For cable well of converter transformer's Control cabinet
	角钢立柱 Angle steel column	L50x50x5，L=750mm	根 Pcs	1	3	3		
	角钢悬架 Angle steel bracket	L40x40x4，L=350mm	套 Set	3	0.89	2.67		
	膨胀螺栓 Expansion bolt	M12x100，热镀锌 M12x100, hot dip galvanizing	套 Set	4				
	钢板 steel plate	80x80x10	套 Set	1	0.5	0.5		
	螺栓 bolt	M12，热镀锌 M12, hot dip galvanizing	套 Set	1				

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR
国家电网 STATE GRID
中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR
East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - E C - 0 4 0 6 - 0 0 3 A

	姓名 Name	日期 Date
设计 Designed	张云燕	2019.01.20
校核 Checked	廖兼明	2019.01.22
审核 Reviewed		
批准 Approved	郑煜	2019.01.24

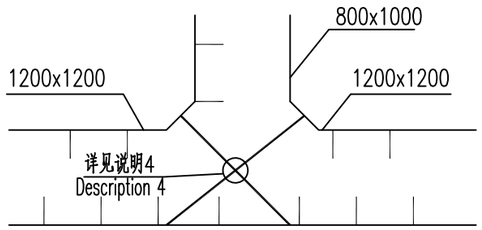
电缆沟支架加工示意图
Detail drawing of cable bracket

比例 Scale:

图号 DWG NO.: 30-B003701S-D0406-003

版本 REV: A

420X594

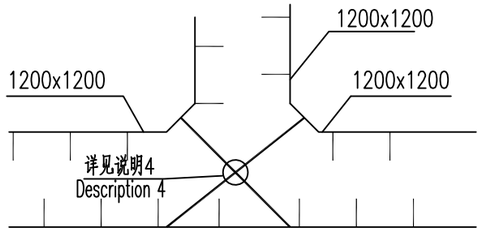


"T"形过渡支架(1)

"T" shaped transition bracket (1)

共2处

2 in total

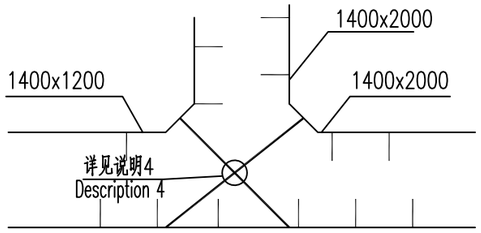


"T"形过渡支架(2)

"T" shaped transition bracket (2)

共19处

19 in total

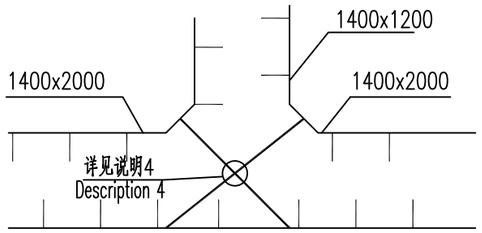


"T"形过渡支架(3)

"T" shaped transition bracket (3)

共2处

2 in total



"T"形过渡支架(4)

"T" shaped transition bracket (4)

共2处

2 in total

说明:

1. 此图仅为示意，具体尺寸及电缆沟内支架布置请施工单位根据现场实际情况制作。
2. 电缆过渡托架的梯边应避免有割坏电缆的风险。
3. 电缆过渡托架角钢立柱与角钢格架采用电焊连接，在托架成型后，将其表面予以热镀锌处理。角钢立柱采用膨胀螺栓固定。
4. 电缆过渡托架横梁交叉处，其中一支横梁应在交叉处断开后进行焊接。
5. 电缆过渡托架的层数与相邻电缆支架层数相等。
6. 电缆过渡支架底层与通长扁钢之间采用M12热镀锌钢制螺栓固定接地。

Description:

1. This figure is only for illustration. The specific dimensions and the arrangement of the brackets in the cable trench should be made by the construction unit according to the actual situation on site.
2. The ladder edge of the cable transition bracket should avoid the risk of cutting the cable.
3. The cable transition bracket angle steel column and angle steel grid are welded by electric welding. After the bracket is formed, the surface is hot-dip galvanized. And the angle steel column is fixed by expansion bolts.
4. At the intersection of the cable transition bracket beams, one of the beams should be welded after being disconnected at the intersection.
5. The number of layers of the cable transition bracket is equal to the number of layers of the adjacent cable bracket .
6. M12 hot-dip galvanized steel bolts shall be used for fixing and grounding between the bottom layer of cable transition bracket and flat steel.

序号 Item number	名称 Name	型号及规范 Model and specification	单位 Unit	数量 Quantity	单重/kg Piece weight/kg	合计/kg Total	总重/kg Gross weight/kg	备注 Remark
1	"T"形过渡支架 "T" shaped transition bracket		套 Set	1			112	共26套 26 in total
	角钢立柱 Angle steel column	L50x50x5，L=1000mm	根 Pcs	4	4	16		
	角钢格架 Angle steel bracket	L50x50x5，L=2000mm	根 Pcs	12	8	96		
	膨胀螺栓 Expansion bolt	M12x100，热镀锌 M12x100, hot dip galvanizing	套 Set	16				
	螺栓 bolt	M12，热镀锌 M12, hot dip galvanizing	套 Set	4				

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C 1 - D D - E C - 0 4 0 6 - 0 0 4 A

	姓名 Name	日期 Date
设计 Designed	张玉鹏	2019.01.20
校核 Checked	傅兼明	2019.01.22
审核 Reviewed		
批准 Approved	郑煜	2019.01.24

电缆过渡支架加工示意图

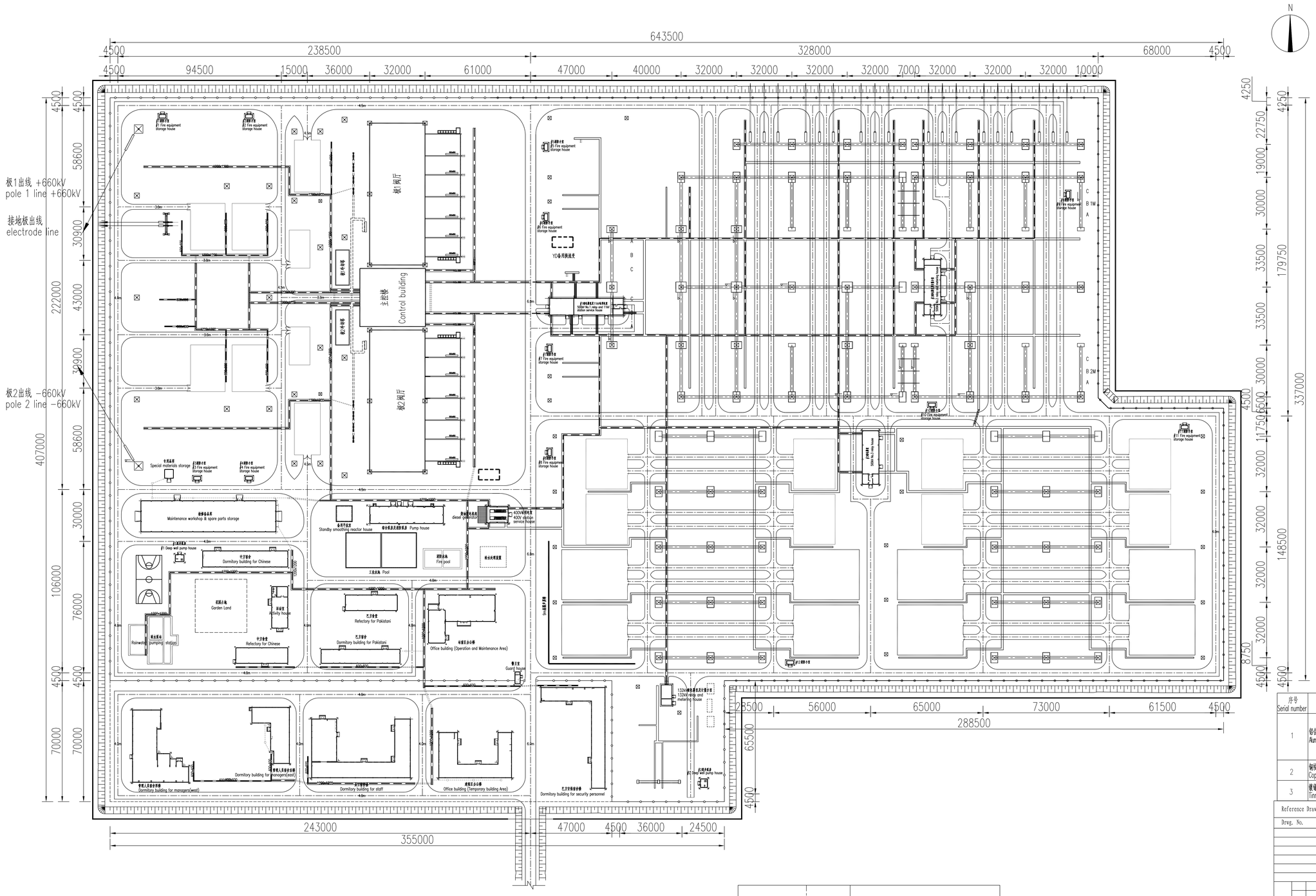
Detail drawing of cable transition bracket

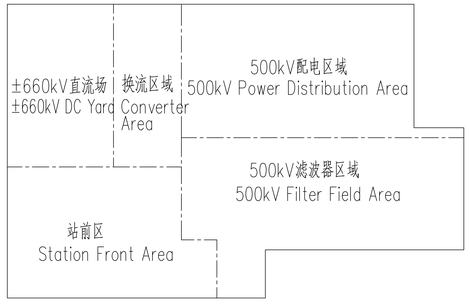
比例 Scale:

图号 DWG NO.: 30-B003701S-D0406-004

版本 REV: A

A3(420x297)





注：1、本图包含全站户外光缆槽盒布置；建筑物内光缆槽盒布置详见各建筑物电缆设施卷册。
"—— —— ——"为光缆槽盒，放置于支架最上层。
2、户内、外光缆槽盒首尾两端采用35mm²铜绞线与电缆沟内通长扁钢螺栓连接，
当光缆槽盒全长大于30m时，应每隔20~30m增加与电缆沟内通长扁钢的连接点。
光缆槽盒连接部位采用TJ-35铜绞线跨接，跨接线随槽盒配供。

Note：1、The volume includes the layout of outdoor cable box.
The layout of cable box inside the building is detailed in the cable facility volume of each building..
The legend "—— —— ——" is cable box. Place cable box on the top layer.
2、Connecting both ends of cable box to the long flat steel in the cable trench with 35mm² copper stranded wire.
When the length of cable box is greater than 30m,connection between the cable box and the long flat steel shall be increase every 20~30m.
Cable box connection parts adopt 35mm² copper stranded wire .The material is provide with cable box.

材料表
Material list

序号 Serial number	名称 Name	型号及规范 Model and specification	单位 Unit	数量 Quantity	备注 Remarks
1	铝合金槽盒及附件 Aluminum alloy box and accessories	150mm×80mm(宽×高)，荷载等级≥D 级 150mm x 80mm (width x height), load grade ≥ D	米 Meter	4,300	附带连接片、伸缩板、接地跨线、螺栓等，敷设在最上层支架 Including connecting piece, expansion plate, ground jumper, bolt, etc Place on the top layer of cable bracket
2	铜绞线 Copper stranded wire	35mm²	米 Meter	250	
3	镀锡铜鼻子 Tinned copper nose	配35mm²铜绞线 Matching 35mm² copper stranded wire	套 Set	350	附带螺栓 With bolt

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT & TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - E C - 0 4 0 6 - 0 0 5 A

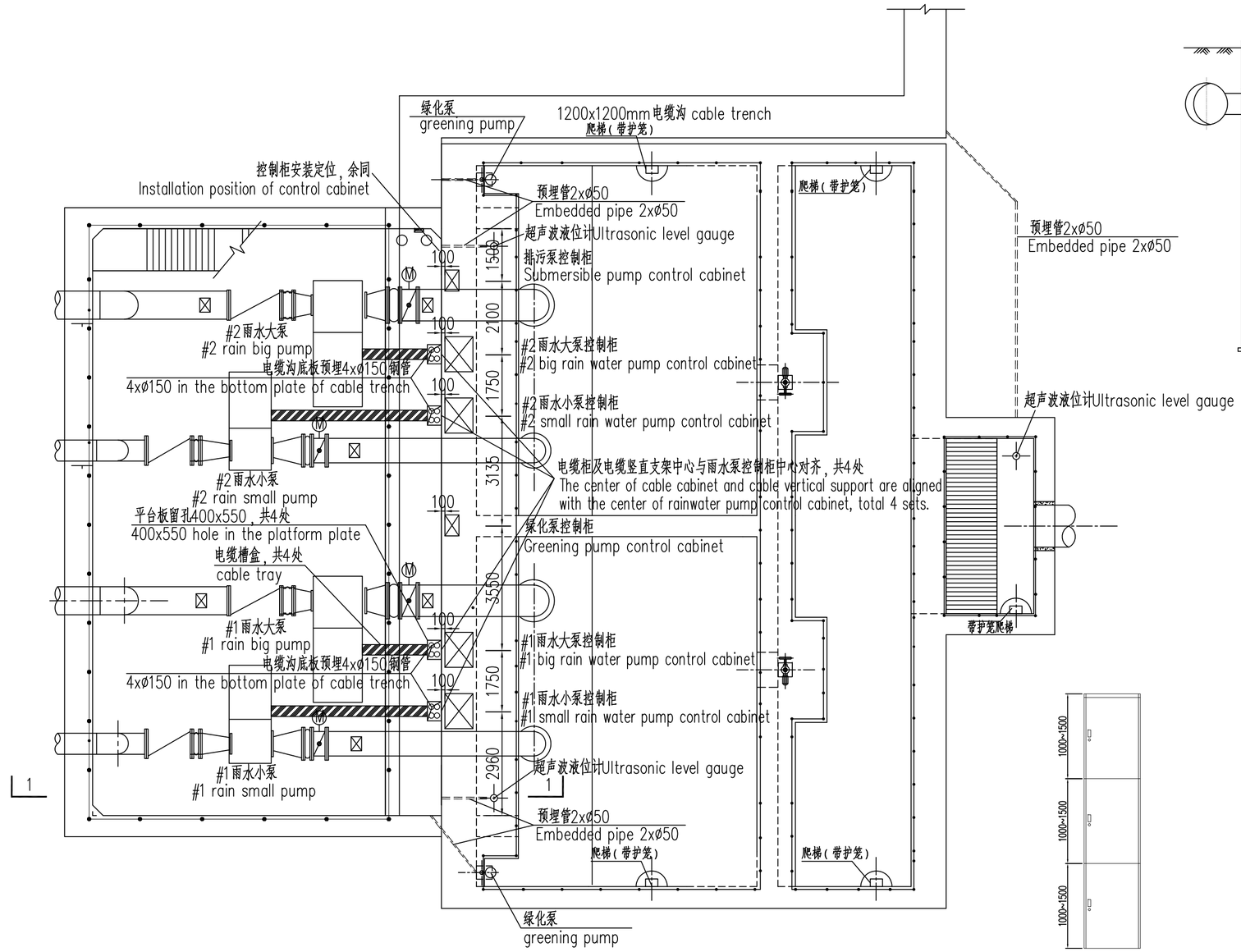
	姓名 Name	日期 Date
设计 Designed	张立德	2019.01.20
校核 Checked	廖基明	2019.01.22
审核 Reviewed		
批准 Approved	郑煜	2019.01.24

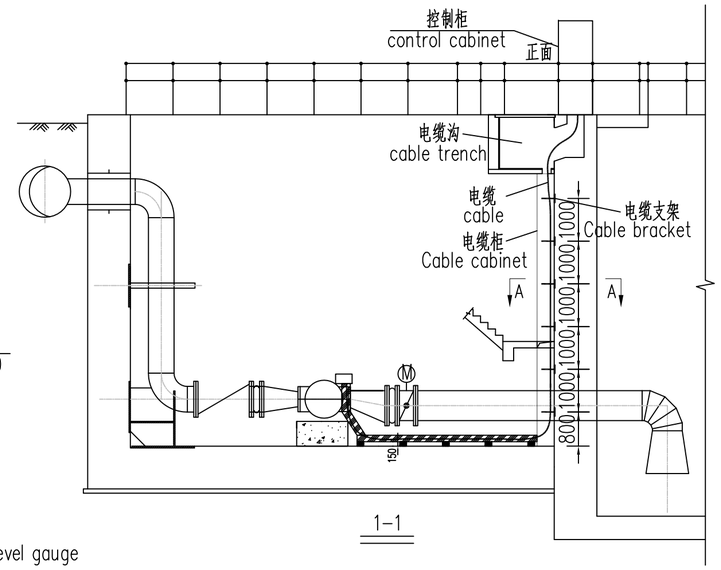
户外电缆沟光缆槽盒布置
Layout drawing of outdoor optical cable box

比例 Scale: 1:800 | 图号 DWG NO.: 30-B003701S-D0406-005 | 版本 REV.: A

专业	会签	日期
通信		
电气二次		

841X1189



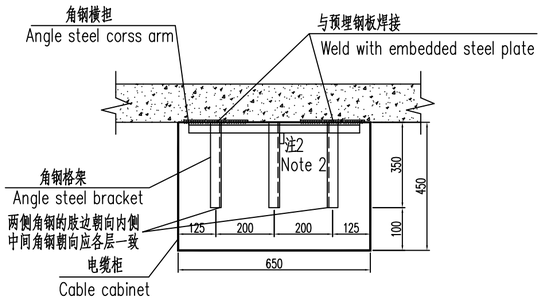


1—1

序号 Serial number	名称 Name	型号及规范 Model and specification	单位 Unit	数量 Quantity	单重/kg Piece weight/kg	合计/kg Total	总重/kg Gross weight/kg	备注 Remark
1	电缆柜 Cable cabinet	650x450x6480(宽x深x高) 650x450x6480 (width x depth x height)	套 Set	4				壳体采用不锈钢 Shell is made of stainless steel
2	电缆支架8 Cable bracket 8		套 Set	1			5.07	用于雨水泵站 For rainwater pump station
	角钢横担 Angle steel corss arm	L50x50x5, L=600mm	根 Pcs	1	2.4	2.4		
	角钢格架 Angle steel bracket	L40x40x4, L=350mm	套 Set	3	0.89	2.67		
	螺栓 bolt	M12, 热镀锌 M12, hot dip galvanizing	套 Set	1				

电缆柜门高1~1.5m
根据现场实际情况调整

Cable cabinet door height 1~1.5m
Adjust according to the actual situation on site



A—A

注: 1、应现场实测后再进行加工,厚度应保证柜体不变形。
2、该处敷设-50x5通长扁钢,材料量开列在D0403卷册内,通长扁钢用热镀锌钢制M12螺栓与每个电缆支架固定,用于电缆支架接地。该通长扁钢上侧与电缆沟内的通长扁钢可靠连接,下侧与雨水泵坑内的辅助接地环网可靠连接。辅助接地环网开列于D0523卷册。
3、角钢支架加工时应保证横架、立柱表面光洁不受破坏;圆角、边缘等处光滑无毛刺和残渣;支架端部应当装设保护套,保护套颜色应全站保持一致。

Note: 1. It shall be processed after on-site measurement, and the thickness shall ensure that the cabinet body is not deformed.
2. -50x5 long flat steel is laid in this place and its material is listed in the Volume D0403. The long flat steel shall be fixed with each cable bracket using M12 hot dip galvanizing bolt for cable bracket grounding. The upper side of this long flat steel is connected with the long flat steel in the cable trench, and the lower side is connected with the auxiliary grounding grid of the rain water pump pit. The auxiliary grounding grid is listed in the Volume D0523.
3. When processing the angle steel bracket, ensure that the surface of the horizontal frame and the column are not damaged; the rounded corners and edges are smooth and free of burrs and debris; A protective cover shall be installed at the end of the bracket, and the color of the protective cover shall be consistent throughout the station.

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - E C - 0 4 0 6 - 0 0 6 A

	姓名 Name	日期 Date
设计 Designed	张云鹏	2019.11.18
校核 Checked	廖兼明	2019.11.18
审核 Reviewed		
批准 Approved	郑煜	2019.11.18
比例 Scale	1:100	图号 DWG NO. 30-B003701S-D0406-006

雨水泵站电缆设施示意图

Detail drawing of cable facilities for rainwater pump station

版本 REV. A

A2(594x420)

序号 Serial number	名 称 Name	型号及规范 Model and specification	单位 Unit	数量 Quantity	重量/kg Quantity/kg	备 注 Remark
	电缆支架 Cable bracket					
1	电缆支架1 Cable bracket 1	600-3-300	套 Set	581	2254.3	用于600x600电缆沟 For 600x600 cable trench
2	电缆支架2 Cable bracket 2	800-4-350	套 Set	110	655.6	用于800x800电缆沟 For 800x800 cable trench
3	电缆支架3 Cable bracket 3	1000-5-350	套 Set	115	879.6	用于800x1000电缆沟 For 800x1000 cable trench
4	电缆支架4 Cable bracket 4	1200-6-350	套 Set	3720	34744.8	用于1200x1200电缆沟 For 1200x1200 cable trench
5	电缆支架5 Cable bracket 5	1200-6-450	套 Set	286	4232.8	用于1400x1200电缆沟 For 1400x1200 cable trench
6	电缆支架6 Cable bracket 6	2000-6-450	套 Set	612	9057.6	用于1400x2000电缆沟 For 1400x2000 cable trench
7	电缆支架7 Cable bracket 7	750-3-350	套 Set	72	444.2	用于换流变汇控柜电缆井 For cable well of converter transformer's Control cabinet
8	电缆支架8 Cable bracket 8		套 Set	24	121.7	用于雨水泵站 For rainwater pump station
9	"T"形过渡支架 "T" shaped transition bracket		套 Set	26	2912	
电缆支架合计 Total Cable bracket		~55.3吨 ~55.3t				
	电缆柜 cable cabinet	650x450x6480(宽x深x高) 650x450x6480 (width x depth x height)	套 Set	4		壳体采用不锈钢 Shell is made of stainless steel
	光缆槽盒 Cable box					
1	铝合金槽盒及附件 Aluminum alloy box and accessories	150mm×80mm(宽x高),荷载等级≥D 级 150mm x 80mm (width x height), load grade ≥ D	米 Meter	4300		敷设在最上层支架 Place on the top layer of cable bracket
2	铜绞线 Copper stranded wire	35mm²	米 Meter	250		
3	镀锡铜鼻子 Tinned copper nose	配35mm²铜绞线 Matching 35mm² copper stranded wire	套 Set	350		附带螺栓 With bolt

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - E C - 0 4 0 6 - 0 0 7 A

	姓名 Name	日期 Date
设计 Designed	张玉燕	2019.01.20
校核 Checked	廖荣明	2019.01.22
审核 Reviewed		
批准 Approved	郑煜	2019.01.24
比例 Scale		图号 DWG NO. 30-B003701S-D0406-007

材料汇总表

Tables of material

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